

SPENCER, IA, USA

WMO: 726500

Lat: 43.164N

Lon: 95.202W

Elev: 408

StdP: 96.52

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14972

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.5	-27.7	0.3	-23.7	-25.1	0.4	-21.1	15.2	-6.7	13.8	-7.4	3.5	280	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	32.9	23.6	31.5	22.8	30.0	21.9	25.5	30.5	24.4	29.5	23.4	28.2	6.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.8	28.7	22.8	18.4	27.6	22.0	17.5	26.7	80.6	30.5	76.0	29.3	71.9	28.0	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.8	10.6	DB	-29.4	35.8	2.8	1.5	-31.4	36.9	-33.1	37.8	-34.7	38.6	-36.7	39.7	(4)
(5)			WB	-29.6	27.3	2.7	1.0	-31.6	28.1	-33.2	28.7	-34.7	29.3	-36.7	30.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-8.4	-6.3	0.9	8.4	15.0	20.7	22.9	21.1	17.4	9.6	1.8	-5.7	(6)	
	DBStd	12.30	7.16	7.11	6.97	5.69	4.91	3.81	3.35	3.18	4.78	5.40	5.99	6.71	(7)	
	HDD10.0	2240	569	458	292	94	12	0	0	0	4	73	252	486	(8)	
	HDD18.3	4110	827	691	541	301	128	19	4	11	73	274	496	744	(9)	
	CDD10.0	1581	0	0	9	46	167	322	399	343	227	61	7	0	(10)	
	CDD18.3	408	0	0	0	2	24	91	146	96	46	4	0	0	(11)	
	CDH23.3	4210	0	0	4	56	300	935	1478	877	489	70	1	0	(12)	
	CDH26.7	1456	0	0	1	16	98	332	564	269	157	18	0	0	(13)	
(14) Wind	WSAvg	5.0	5.3	5.4	5.6	6.1	5.6	4.6	3.7	3.5	4.2	4.9	5.2	5.2	(14)	
(15) Precipitation	PrecAvg	743	14	16	40	80	103	118	94	97	74	55	33	20	(15)	
	PrecMax	1092	30	50	87	166	165	263	175	247	327	161	101	47	(16)	
	PrecMin	552	1	2	0	22	29	41	15	31	5	12	1	3	(17)	
	PrecStd	130	8	11	24	37	39	60	42	54	65	37	27	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	13.0	22.9	29.2	32.8	34.5	35.0	33.7	32.7	29.3	21.4	12.4	(19)	
		MCWB	5.0	9.2	14.0	17.1	20.3	23.1	25.1	24.5	21.9	18.4	13.0	9.4	(20)	
	2%	DB	5.0	8.2	18.3	24.4	29.2	32.2	32.9	31.4	30.8	25.0	17.3	8.0	(21)	
		MCWB	2.5	5.7	12.2	14.6	18.4	22.2	24.8	23.7	21.5	16.4	11.6	4.9	(22)	
	5%	DB	2.6	5.1	14.2	21.4	26.9	30.3	31.6	29.4	28.3	21.9	13.8	4.8	(23)	
		MCWB	0.9	3.0	10.0	12.9	17.4	21.2	24.1	22.6	19.9	15.2	9.1	2.5	(24)	
	10%	DB	1.0	2.7	11.0	18.2	24.0	28.3	29.9	27.8	26.3	18.9	11.0	2.7	(25)	
		MCWB	-0.1	1.1	7.8	11.5	16.0	20.4	22.9	21.5	18.9	13.2	7.4	1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	9.9	16.0	18.0	22.2	25.3	27.6	26.3	24.0	20.9	14.8	10.5	(27)	
		MCDB	9.2	13.5	19.7	25.9	28.9	31.2	32.1	31.4	30.0	25.9	19.5	12.6	(28)	
	2%	WB	2.8	5.7	13.0	15.8	20.2	23.9	26.0	24.9	22.7	18.0	12.1	5.2	(29)	
		MCDB	4.7	8.2	17.5	22.7	26.8	29.8	30.8	29.7	28.1	22.1	16.1	8.2	(30)	
	5%	WB	1.1	3.2	10.5	14.1	18.7	22.7	25.0	23.7	21.6	16.0	9.9	2.9	(31)	
		MCDB	2.4	5.0	14.1	19.8	24.6	28.1	29.9	28.0	26.3	20.5	13.4	4.7	(32)	
	10%	WB	-0.2	1.3	7.6	12.2	17.2	21.6	24.0	22.6	20.3	13.9	7.4	1.3	(33)	
		MCDB	0.9	2.5	11.0	16.8	22.2	26.9	28.8	26.5	24.6	18.5	10.8	2.6	(34)	
(35) Mean Daily Temperature Range	MDBR		9.7	9.8	10.2	12.6	12.4	12.0	12.0	12.1	13.8	12.8	11.3	9.4	(35)	
	5% DB	MCDBR	11.8	11.8	16.1	18.0	16.3	14.0	12.7	13.2	15.2	16.8	16.3	12.4	(36)	
		MCWBR	9.6	9.3	10.4	9.7	7.6	6.4	6.2	6.9	7.3	9.2	10.6	9.6	(37)	
	5% WB	MCDBR	10.7	11.0	15.3	15.3	13.6	11.9	11.6	11.7	12.7	13.9	14.7	11.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.271	0.287	0.314	0.363	0.389	0.405	0.416	0.409	0.367	0.330	0.290	0.273	(40)	
	taud		2.433	2.405	2.428	2.321	2.318	2.332	2.284	2.288	2.375	2.467	2.503	2.459	(41)	
	Ebn at Noon		876	918	931	902	884	868	853	847	860	855	847	840	(42)	
	Edh at Noon		74	90	101	122	126	124	129	125	106	85	69	66	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.88	2.96	3.83	4.76	5.57	6.18	6.56	5.46	4.40	2.97	2.02	1.50	(44)	
	RadStd		0.19	0.27	0.35	0.43	0.40	0.46	0.30	0.39	0.29	0.38	0.14	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page